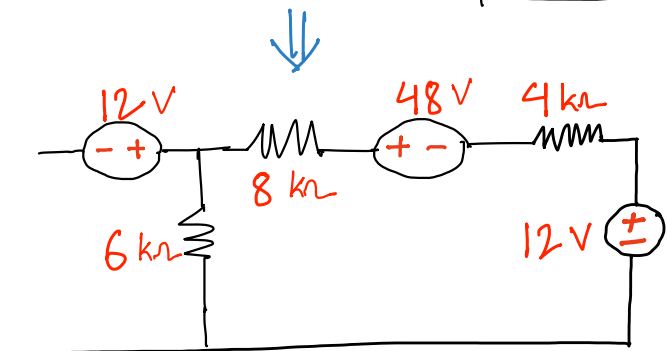
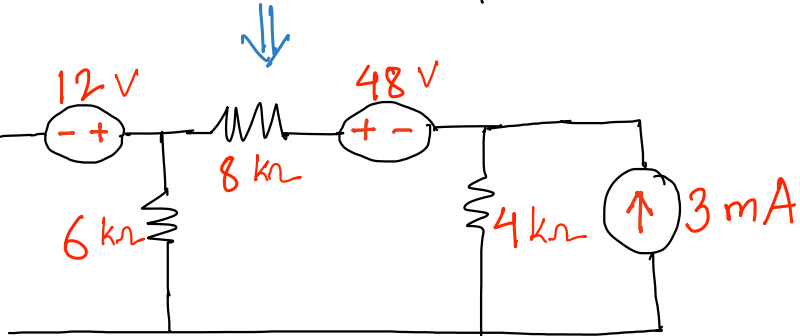
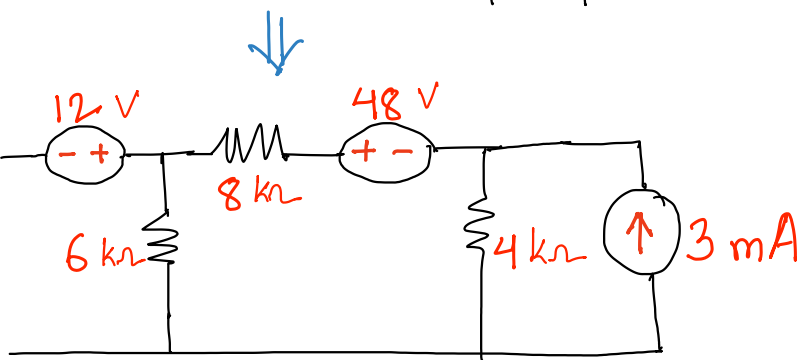
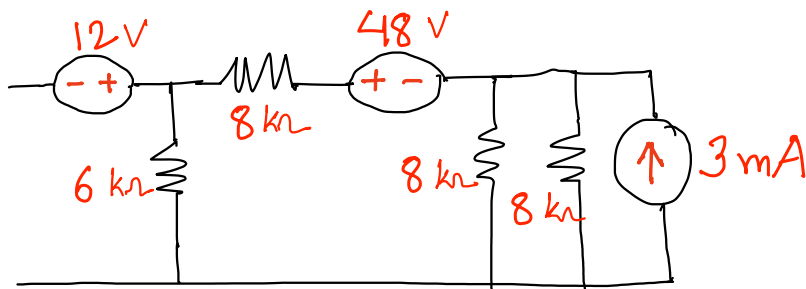
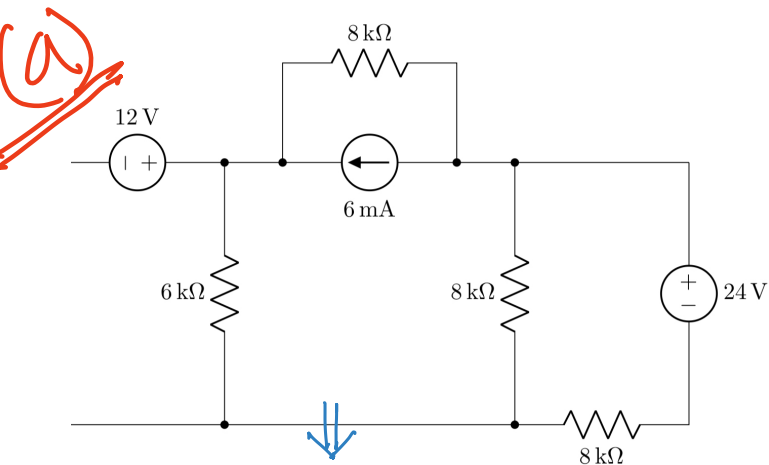


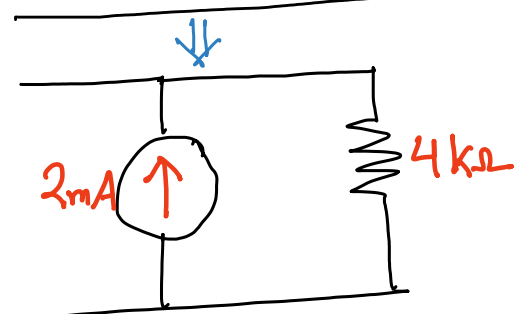
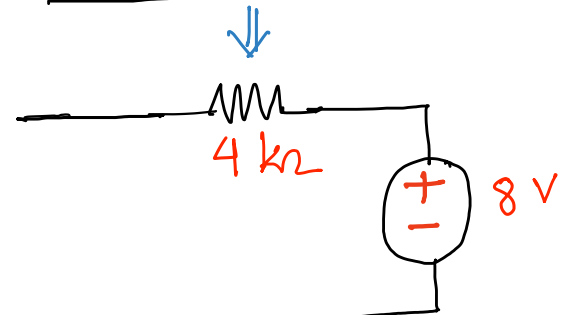
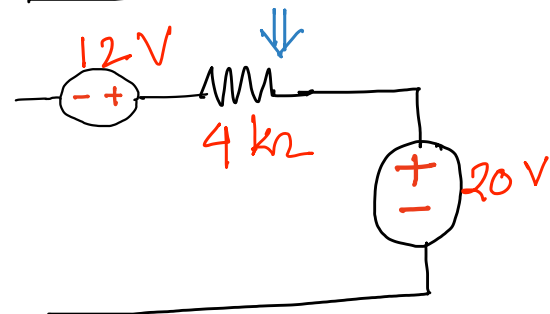
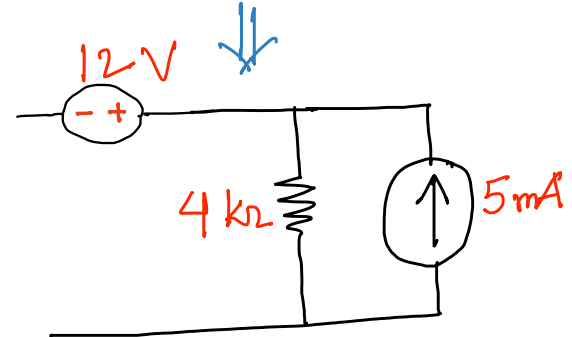
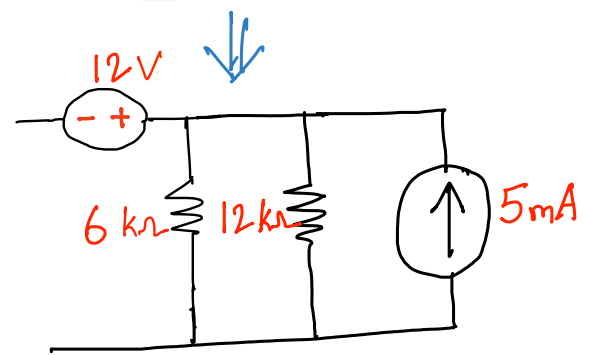
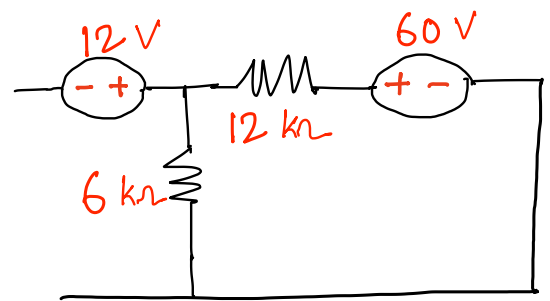
Set B

Question - 1

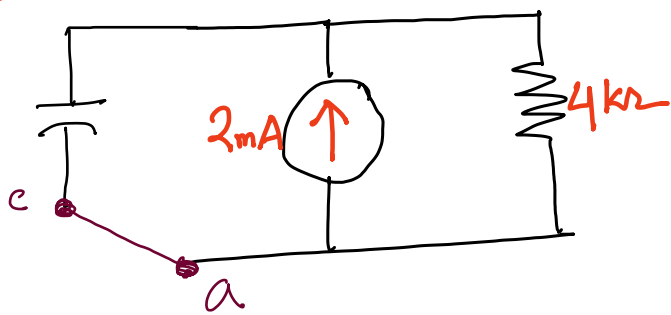


$$I_o = 2 \text{ mA}$$

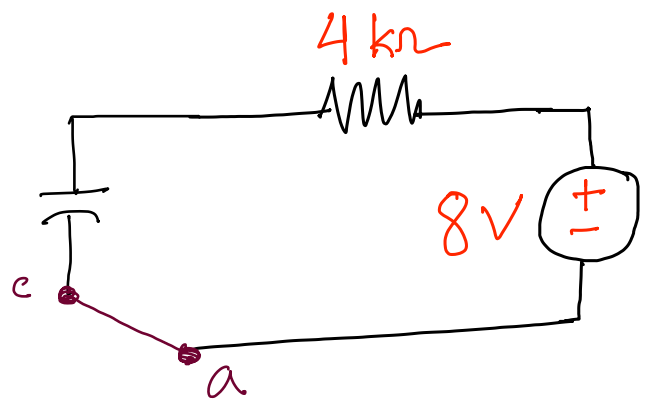
$$R_o = 4 \text{ k}\Omega$$



(b) when $t < 0$:

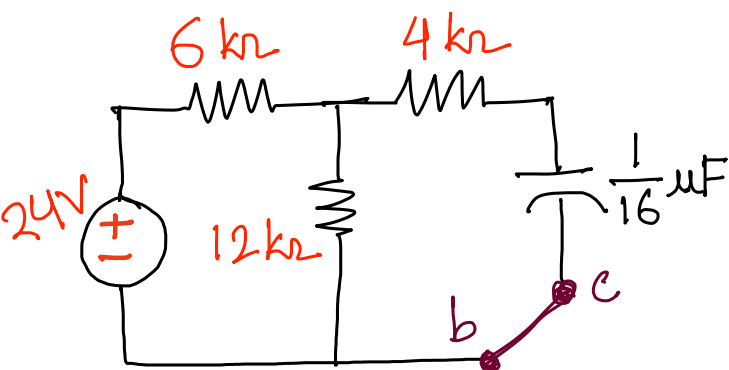


$\Rightarrow$



after a long time, the capacitor will be fully charged to $\rightarrow V_c(0) = 8V = V_{ini}$
initial voltage

when $t > 0$:

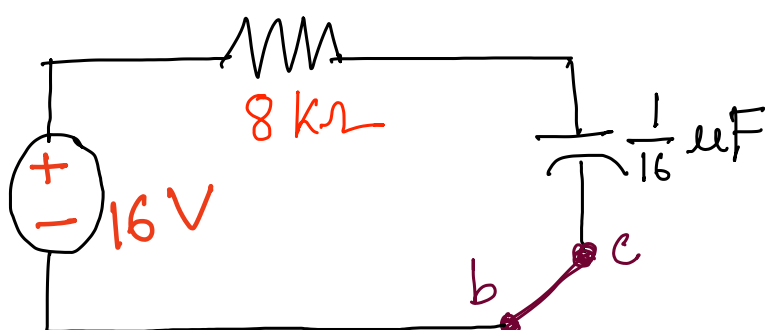
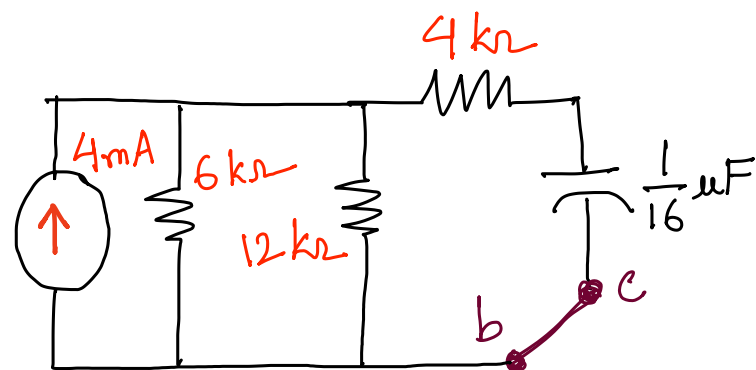


Again, after a long time, the capacitor will be fully charged to $\rightarrow$

$$V_c(\infty) = 16V = V_{final}$$

final voltage

at $t > 0$



$$\begin{aligned} \tau &= R_{eq} \cdot C \\ &= (8 \times 10^3) \cdot \left(\frac{1}{16} \times 10^{-6}\right) \\ &= 5 \times 10^{-4} \text{ Sec} \\ \tau &= 0.5 \text{ ms} \end{aligned}$$

$$V_c(t) = V_{final} + [V_{ini} - V_{final}] e^{-t/\tau}$$

$$= 16 + [8 - 16] e^{-\frac{t}{5 \times 10^{-4}}} \text{ volts}$$

$$V_c(t) = 16 - 8 e^{-2000t} \text{ volts}$$

(c)

$$i_c(t) = C \cdot \frac{dV_c}{dt}$$

$$= \left(\frac{1}{16} \times 10^{-6} \right) \times \frac{d}{dt} [16 - 8 e^{-2000t}]$$

$$= 0 - (-2000) \times \frac{8}{16} \times 10^{-6} \cdot e^{-2000t}$$

$$= 0.001 e^{-2000t} \text{ A}$$

$$i_c(t) = 1 e^{-2000t} \text{ mA}$$

Question-2

(a) from graph,

$$P_{R_L(\text{Max})} = 8 \text{ W}$$

$$R_L = 2 \Omega$$

$$\therefore R_{TH} = R_L = 2 \Omega ;$$

$$P_{R_L \text{ Max}} = \frac{V_{TH}^2}{4 R_{TH}}$$

$$\Rightarrow V_{TH} = \sqrt{8 \times 4 \times 2}$$

$$\therefore V_{TH} = 8 \text{ volts}$$

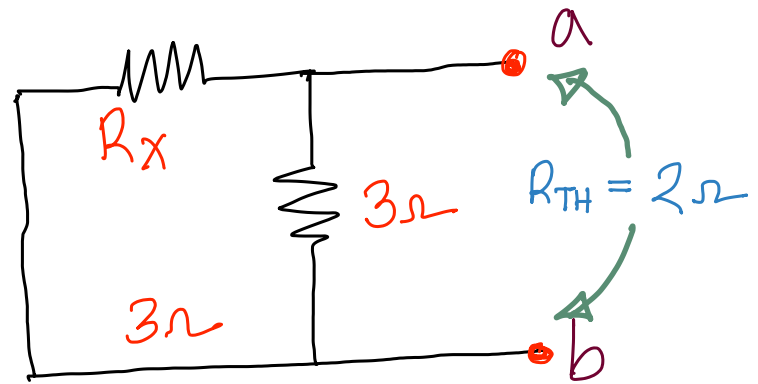
(i) Figure out $R_x \rightarrow$

Killing all independent sources $\rightarrow$

$$R_{TH} = (R_x + 3) \parallel 3$$

$$2 = \frac{3 \times (R_x + 3)}{3 + R_x + 3}$$

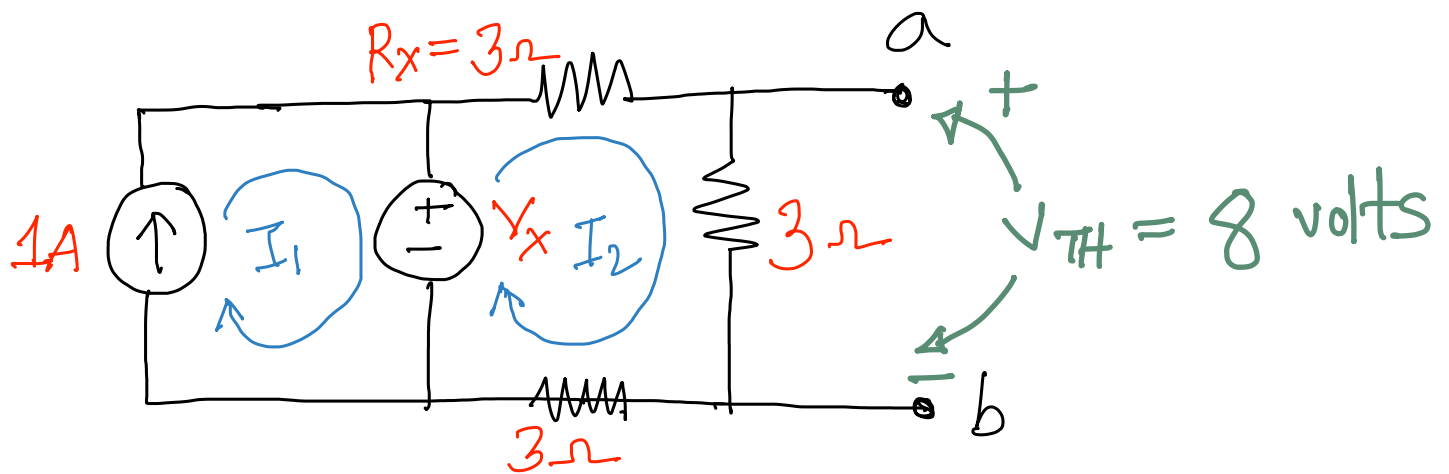
$$\Rightarrow 2R_x + 12 = 3R_x + 9$$



$$\Rightarrow R_x = 12 - 9$$

$$\therefore R_x = 3\Omega$$

(ii) Figure out $V_x \rightarrow$



$$I_1 = 1A$$

$$\text{Here, } V_{TH} = 3I_2 \Rightarrow I_2 = \frac{V_{TH}}{3} = \frac{8}{3}$$
$$\therefore I_2 = \frac{8}{3}$$

KVL in Mesh-2 $\Rightarrow$

$$-V_x + 3I_2 + 3I_2 + 3I_2 = 0$$

$$\Rightarrow 9I_2 = V_x$$

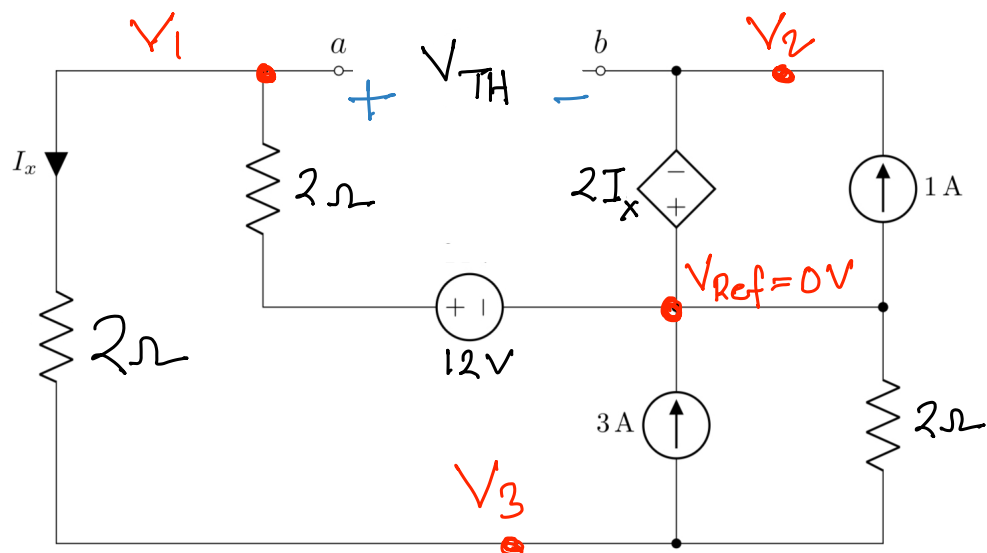
$$\Rightarrow V_x = 9 \times \frac{8}{3}$$

$$\therefore V_x = 24 \text{ volts}$$

(b)

Figure out V_{TH}

Using Nodal Analysis $\rightarrow$



$$\frac{V_1 - 12}{2} + \frac{V_1 - V_3}{2} = 0$$
$$\Rightarrow 2V_1 - V_3 = 12 \dots\dots \textcircled{i}$$

$$\left[\text{here, } I_x = \frac{V_1 - V_3}{2} \right]$$

$$\frac{V_3 - V_1}{2} + 3 + \frac{V_3 - 0}{2} = 0$$

$$\Rightarrow 2V_3 - V_1 = -6$$

$$\therefore V_1 - 2V_3 = 6 \dots\dots \textcircled{iii}$$

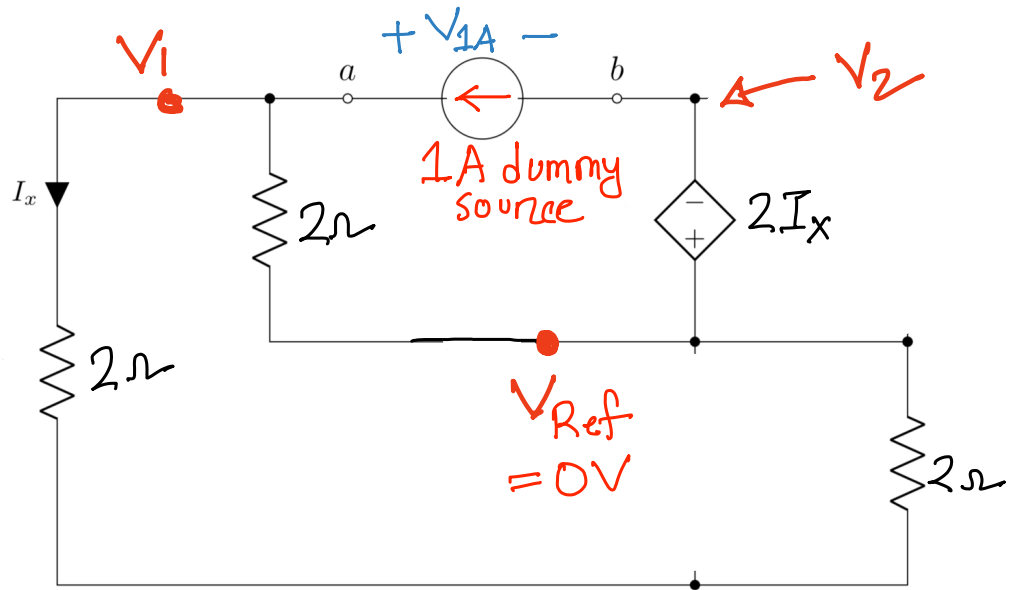
Solving $\textcircled{i}$, $\textcircled{ii}$ and $\textcircled{iii} \Rightarrow$

$$0 - V_2 = 2I_x$$
$$\Rightarrow V_2 = 2(-I_x)$$
$$\Rightarrow V_2 = 2 \cdot \left(\frac{V_3 - V_1}{2} \right)$$
$$\Rightarrow V_2 = V_3 - V_1$$
$$\Rightarrow V_1 + V_2 - V_3 = 0 \dots\dots \textcircled{ii}$$

$$\begin{cases} V_1 = 6 \text{ volts} \\ V_2 = -6 \text{ volts} \\ V_3 = 0 \text{ volts} \end{cases}$$

$$V_{TH} = V_1 - V_2 = 6 - (-6) = 12 \text{ volts}$$

Figure out R_{TH}



Firstly,

$$\frac{V_1 - 0}{2} + \frac{V_1 - 0}{(2+2)} - 1 = 0$$

$$\Rightarrow \frac{2V_1 + V_1}{4} = 1$$

$$\left[I_x = \frac{V_1 - 0}{(2+2)} \right]$$

$$\Rightarrow 3V_1 = 4$$

$$\Rightarrow V_1 = 4/3 \text{ volts}$$

Now,

$$0 - V_2 = 2 \cdot I_x$$

$$\Rightarrow V_2 = 2 \cdot (-I_x)$$

$$\Rightarrow V_2 = 2 \cdot \left(\frac{0 - V_1}{2+2} \right)$$

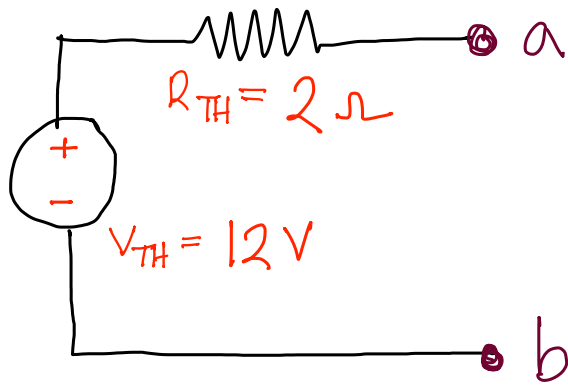
$$\Rightarrow V_2 = 2 \cdot \left(\frac{0 - \frac{4}{3}}{4} \right)$$

$$\therefore V_2 = -\frac{2}{3} \text{ volts}$$

$$\begin{aligned} \therefore V_{1A} &= V_1 - V_2 \\ &= \frac{4}{3} - \left(-\frac{2}{3} \right) \\ &= 2 \text{ volts} \end{aligned}$$

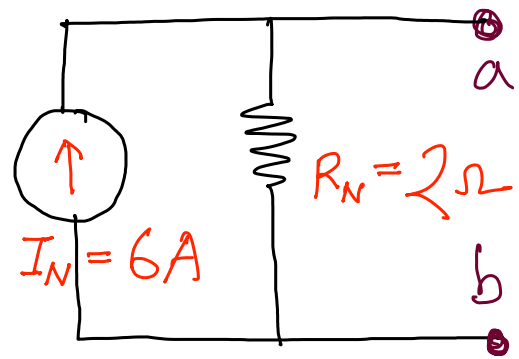
$$\begin{aligned} \therefore R_{TH} &= \frac{V_{1A}}{1A} \\ &= \frac{2}{1} \end{aligned}$$

$$\therefore R_{TH} = 2 \Omega$$

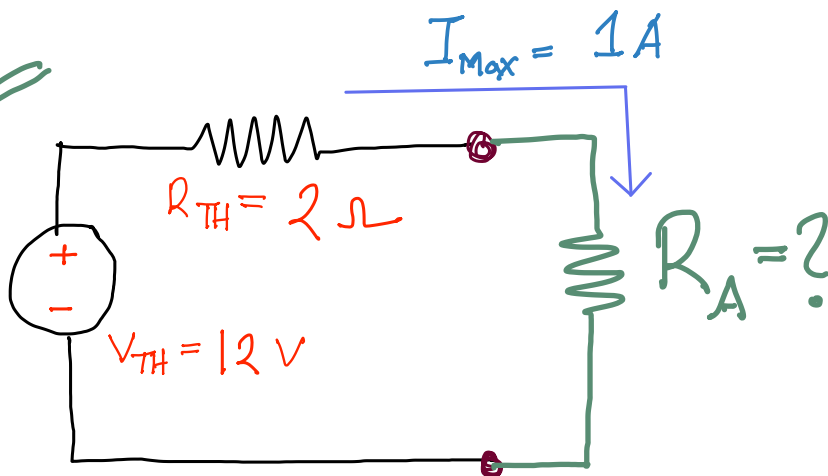
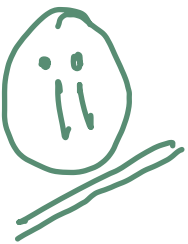


Thevenin's Equivalent circuit

and,



Norton's Equivalent circuit



Thevenin's Equivalent circuit

let's replace Ammeter with ^{min} Resistance ' R_A '

$$I_{Max} = \frac{V_{TH}}{R_{TH} + R_A} \Rightarrow 1 = \frac{12}{2 + R_A}$$

$$\therefore R_A = 10\Omega$$

Question-3

$$\frac{V - V_1}{V_1 - V_2} = \frac{I - I_1}{I_1 - I_2}$$

$$\Rightarrow \frac{V - 2}{2 - 6} = \frac{I + 0.5}{-0.5 - 0.5}$$

$$\Rightarrow \frac{V - 2}{-4} = \frac{I + 0.5}{-1}$$

$$\Rightarrow V - 2 = 4I + 2$$

$$\Rightarrow 4I = V - 4$$

$$\Rightarrow \boxed{I = \frac{V}{4} - 1}$$

$\Rightarrow$ I-V characteristics equation of the linear circuit 'X'

Now,

when $V = 10$ volts

$$\rightarrow I = \frac{10}{4} - 1$$

$$\therefore \boxed{I = 1.5 \text{ A}}$$

when $I = 2 \text{ A}$

$$\rightarrow 2 = \frac{V}{4} - 1$$

$$\therefore \boxed{V = 12 \text{ volts}}$$

(a)

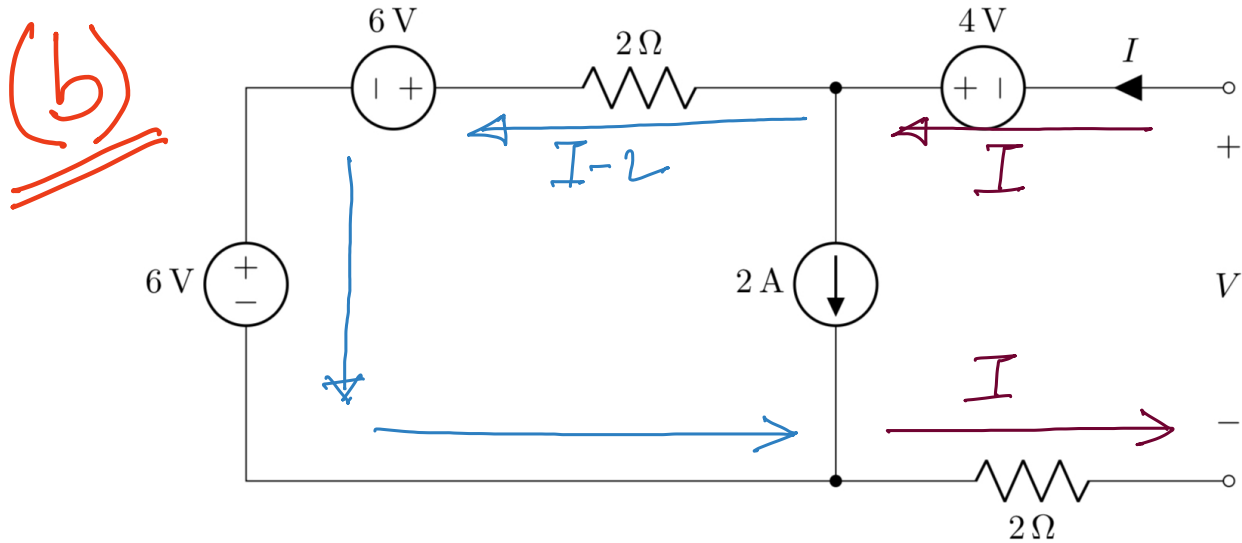
V (volt)	2	6	10	
I (A)	-0.5	0.5		2.0

$$(V_1, I_1) = (2, -0.5)$$

$$(V_2, I_2) = (6, 0.5)$$

completed
table:

V (volt)	2	6	10	12
I (A)	-0.5	0.5	1.5	2.0



KVL at the bigger loop:

$$-V + (-4) + 2 \times (I-2) + 6 + 6 + 2 \times I = 0$$

$$\Rightarrow 4I = V - 4$$

$$\therefore \boxed{I = \frac{V}{4} - 1}$$

The following I - V characteristics equation matches with the I - V equation of circuit 'X'

Hence, they are both equivalent.

(Showed)